CONTRACTION OF AN EXCEPTIONAL CURVE OF THE FIRST KIND PRESERVING PROJECTIVITY AND QUASI-PROJECTIVITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the contraction over an affine open of the base together with the ample module produced there, the glueing of the contracted piece with the complement of the curve and the verification of properness by localness on the base, and the glueing of the invertible modules yielding relative ampleness followed by the reduction of the quasi-projective case to a projective model.

Editorial note. The label lemma-contract-when-quasi-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-contract-when-quasi-projective –start 4985 –end 5047 –local-id 28082 returned RESULT=OK (L28082 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Contraction over an affine base with an ample module on the target. Same chapter, the lemma producing a contraction of an exceptional curve of the first kind over an affine base together with an ample invertible module on the target whose pullback is a twist of the given module. Supplies the contraction over an affine open containing the image of the curve and the ample invertible module on the target.
- Characterization of projectivity and of quasi-projectivity through relative ampleness. Morphisms, the lemmas relating a projective morphism to properness plus the existence of a relatively ample invertible module, and the lemma writing a quasi-projective scheme as an open subscheme of a projective one. Supply the description of the two hypotheses and the embedding used in the second part.
- Localness of properness on the base. Morphisms, the lemma that properness can be checked after restricting to the members of an open covering of the base. Supplies the conclusion that the glued morphism is proper over the base.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0C2M: sources/stacks-resolve/pinned-original-source.tex, source lines 4883–4883. Prerequisite pointer invoked at proof line(s) 4999; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 01VW: sources/stacks-chow/tags.txt, source lines 2461–2461. Prerequisite pointer invoked at proof line(s) 5007; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01VJ: sources/stacks-chow/tags.txt, source lines 2448–2448. Prerequisite pointer invoked at proof line(s) 5032; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BCL: sources/stacks-chow/tags.txt, source lines 14284–14284. Prerequisite pointer invoked at proof line(s) 5006; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01W2: sources/stacks-chow/tags.txt, source lines 2467–2467. Prerequisite pointer invoked at proof line(s) 5023; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07RM: sources/stacks-chow/tags.txt, source lines 9872–9872. Prerequisite pointer invoked at proof line(s) 5038; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a morphism of finite type. Let $E \subset X$ be an exceptional curve of the first kind which is in a fibre of f.*

(1) *If X is projective over S, then there exists a contraction $X \to X'$ of E and X' is projective over S.*
(2) *If X is quasi-projective over S, then there exists a contraction $X \to X'$ of E and X' is quasi-projective over S.*

Proof. Both cases follow from Lemma 0C2M using standard results on ample invertible modules and (quasi-)projective morphisms.

Proof of (1). Projectivity of f means that f is proper and there exists an f-ample invertible module $\mathcal{L}$, see Morphisms, Lemma 0BCL and Definition 01VW. Let $U \subset S$ be an affine open containing the image of E. By Lemma 0C2M there exists a contraction $c : f^{-1}(U) \to V'$ of E and an ample invertible module $\mathcal{N}'$ on V' whose pullback to $f^{-1}(U)$ is equal to $\mathcal{L}(nE)|_{f^{-1}(U)}$. Let $v \in V'$ be the closed point such that c is the blowing up of v. Then we can glue V' and $X \setminus E$ along $f^{-1}(U) \setminus E = V' \setminus \{v\}$ to get a scheme X' over S. The morphisms c and $\mathrm{id}_{X \setminus E}$ glue to a morphism $b : X \to X'$ which is the contraction of E. The inverse image of U in X' is proper over U. On the other hand, the restriction of $X' \to S$ to the complement of the image of v in S is isomorphic to the restriction of $X \to S$ to that open. Hence $X' \to S$ is proper (as being proper is local on the base by Morphisms, Lemma 01W2). Finally, $\mathcal{N}'$ and $\mathcal{L}|_{X \setminus E}$ restrict to isomorphic invertible modules over $f^{-1}(U) \setminus E = V' \setminus \{v\}$ and hence glue to an invertible module $\mathcal{L}'$ over X'. The restriction of $\mathcal{L}'$ to the inverse image of U in X' is ample because this is true for $\mathcal{N}'$. For affine opens of S avoiding the image of v, we see that the same is true because

it holds for $\mathcal{L}$. Thus $\mathcal{L}'$ is $(X' \to S)$-relatively ample by Morphisms, Lemma 01VJ and (1) is proved.

Proof of (2). We can write X as an open subscheme of a scheme $\overline{X}$ projective over S by Morphisms, Lemma 07RM. By (1) there is a contraction $b : \overline{X} \to \overline{X}'$ and $\overline{X}'$ is projective over S. Then we let $X' \subset \overline{X}$ be the image of $X \to \overline{X}'$; this is an open as b is an isomorphism away from E. Then $X \to X'$ is the desired contraction. Note that X' is quasi-projective over S as it has an S-relatively ample invertible module by the construction in the proof of part (1). $\square$